JMMMU: A Japanese Massive Multi-discipline Multimodal Understanding Benchmark for Culture-aware Evaluation

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Abstract

Existing multimodal benchmarks for evaluating Large Multimodal Models (LMMs) are predominantly English-centric, leaving significant gaps in rigorous, culture-aware evaluation for other languages. We introduce **JMMMU** (**J**apanese **M**assive **M**ulti-discipline **M**ultimodal **U**nderstanding), the first large-scale, expert-level Japanese multimodal benchmark designed for culture-aware evaluation. JMMMU contains 1,320 questions paired with 1,118 images across 28 subjects, and is structured into two complementary subsets: (1) a *culture-agnostic* (CA) subset of 720 questions translated from the English MMMU benchmark, enabling direct English–Japanese performance comparisons, and (2) a *culture-specific* (CS) subset of 600 newly created questions covering authentically Japanese domains (Japanese Art, Heritage, History, and World History). This design allows JMMMU to disentangle language proficiency from cultural understanding. Comprehensive evaluation of 16 open-source and 3 proprietary LMMs reveals significant performance gaps: the best open-source model achieves only 40.5%, while the best proprietary model (GPT-4o) reaches 58.6%. Even on the CA subset, models show an average -4.3% drop when switching from English to Japanese. On the CS subset, GPT-4o outperforms Claude 3.5 Sonnet by $+15.7\%$, underscoring that strong Japanese language handling does not imply cultural understanding. JMMMU establishes a rigorous foundation for advancing multilingual and culturally-aware multimodal AI systems.

1 Introduction

Large Multimodal Models (LMMs) have achieved impressive performance on a wide range of vision-language tasks (Liu et al., 2023; Alayrac et al., 2022; Chen et al., 2024; Li et al., 2024). However, the evaluation of these models has been predominantly focused on English, with benchmarks such as MMMU (Yue et al., 2024a), MMBench (Liu et al., 2024b), MathVista (Lu et al., 2024), VQA (Antol et al., 2015), ChartQA (Masry et al., 2022), and DocVQA (Mathew et al., 2021) setting the standards for measuring expert-level and common-sense visual understanding. As LMMs are deployed globally, there is a growing need for comprehensive, culturally grounded evaluation beyond English—yet multilingual multimodal benchmarks remain scarce.

Japanese represents a particularly important yet underserved case. While several Japanese LMM benchmarks exist (SakanaAI, 2024b; Turing, 2024; SakanaAI, 2024a,c; Inoue et al., 2024b), they share critical limitations: all are small in scale (50–500 questions), restricted to common-sense reasoning, and largely ignore expert-level knowledge or culturally grounded content. Some rely on direct translation from English benchmarks without adapting for Japanese cultural context, making it impossible to determine whether a model genuinely understands Japanese culture or merely exhibits language transfer. Furthermore, no existing Japanese benchmark offers a systematic means to disentangle language ability from cultural competence.

To address these gaps, we introduce **JMMMU**, a Japanese Massive Multi-discipline Multimodal Understanding benchmark designed for culture-aware evaluation of LMMs. JMMMU contains 1,320 questions paired with 1,118 images spanning 28 subjects, requiring expert-level, college-grade reasoning. A key architectural decision of JMMMU is its two-part structure: a *culture-agnostic* (CA) subset of 720 questions translated and localized from MMMU (Yue et al., 2024a), and a *culture-specific* (CS) subset of 600 questions in newly created subjects (Japanese Art, Japanese Heritage, Japanese History, and World History aligned to Japanese curricula). The CA subset enables clean apples-to-apples comparisons between English and Japanese performance on identical content, while the CS

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subset probes genuine cultural knowledge.

Our key contributions are:

- We introduce JMMMU, the **first large-scale, expert-level Japanese multimodal benchmark** (1,320 questions; 1,118 images) spanning 28 subjects with both culture-agnostic and culture-specific evaluation.
- We provide a **rigorous construction pipeline** including GPT-4o-assisted translation with human expert refinement, currency normalization, and manual image localization (English text → Japanese), with 19 expert native annotators ensuring quality.
- We enable **clean English–Japanese comparisons** via the CA subset translated from MMMU, and introduce novel CS subjects that reflect authentic Japanese cultural knowledge.
- We deliver **comprehensive evaluation and analysis** of 19 LMMs, including language-vs-culture disentanglement, translation ablation studies, and a four-category error taxonomy to guide multilingual LMM development.

2 Related Work

2.1 Multimodal Benchmarks

The field of multimodal understanding has been shaped by a series of increasingly challenging benchmarks. Early VQA benchmarks (Antol et al., 2015) evaluated basic image-question answering. More recent benchmarks test specialized capabilities: DocVQA (Mathew et al., 2021) for document understanding, ChartQA (Masry et al., 2022) for chart reasoning, and MathVista (Lu et al., 2024) for mathematical reasoning. MMMU (Yue et al., 2024a) marked a significant advance by introducing 11.5K expert-level, multi-discipline questions requiring college-level reasoning across 30 subjects, and has become a de facto standard for evaluating frontier LMMs. MMMU-Pro (Yue et al., 2024c) further raised difficulty by increasing the number of options and introducing vision-only questions. MMBench (Liu et al., 2024b) evaluates a broader range of visual abilities using a structured taxonomy. Our work extends this line of research to Japanese, building directly on the MMMU framework while adding culturally grounded content.

2.2 Multilingual and Japanese LMM Benchmarks

The need for multilingual evaluation has led to a number of non-English benchmarks. CMMMU (Zhang et al., 2024) adapts MMMU to Chinese, providing college-level evaluation in Chinese with cultural localization. However, for Japanese, the benchmark landscape remains limited. As summarized in Table 1, existing Japanese LMM benchmarks (SakanaAI, 2024b; Turing, 2024; SakanaAI, 2024a,c; Inoue et al., 2024b) are uniformly small (50–500 questions), test only common-sense rather than expert knowledge, and most do not evaluate culturally specific content. JMMMU fills this gap as the first Japanese expert-level multimodal benchmark with systematic culture-aware design.

Japanese poses distinctive structural challenges for multimodal evaluation that set it apart from other languages studied in the benchmark literature. Its three interwoven writing systems—hiragana, katakana, and kanji—impose non-trivial tokenization demands on model vocabularies trained predominantly on Latin-script corpora. Domain terminology in fields such as classical art, law, and medicine frequently lacks direct English cognates, making translation-based evaluation unreliable for expert-level tasks. Moreover, Japanese discourse relies heavily on implicit context and honorific register, meaning that even grammatically correct machine translations may alter the pragmatic difficulty of a question in ways that are invisible to automated quality metrics. These properties collectively imply that Japanese-language benchmarking requires native expert oversight at every stage of construction, not merely post-hoc review.

Beyond Japanese, efforts to evaluate multilingual multimodal models have highlighted that translation alone is insufficient for culture-aware benchmarking. CMMMU (Zhang et al., 2024) demonstrates that even high-quality Chinese translations of MMMU reveal substantial performance gaps relative to native English evaluation. The broader lesson is that languages with distinct scripts and cultural contexts—such as Japanese, with its mixed kanji-kana writing system—introduce evaluation challenges that go beyond vocabulary or grammar differences. Models must handle specialized domain terminology with no direct English cognate, culturally loaded visual cues, and expert-level questions that simultaneously test language comprehension and cultural knowledge. JMMMU addresses

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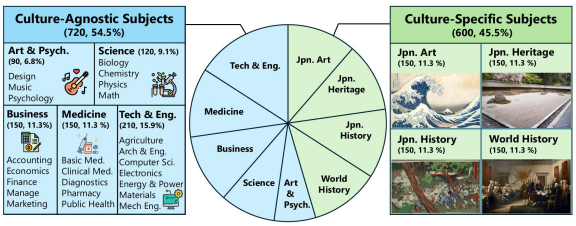

Figure 1: **Overview of the JMMMU dataset.** JMMMU consists of 1,320 questions across 28 subjects, split into a culture-agnostic (CA) subset of 720 questions translated from MMMU and a culture-specific (CS) subset of 600 newly created questions in four Japanese-oriented subjects.

this by cleanly separating language-induced difficulty (CA subset) from cultural knowledge requirements (CS subset), enabling fine-grained attribution of model failures that purely translation-based benchmarks cannot provide.

2.3 Large Multimodal Models

Rapid progress in LMMs has been driven by instruction-tuned models connecting visual encoders with large language models. Open-source models include the LLaVA family (Liu et al., 2023, 2024a; Li et al., 2024), InternVL (Chen et al., 2023, 2024), Phi-3-Vision (Abdin et al., 2024), CogVLM2 (Hong et al., 2024), xGen-MM (Xue et al., 2024), Mantis (Jiang et al., 2024), Idefics (Awadalla et al., 2023), and Pangea (Yue et al., 2024b). Japanese-specialized open-source models include LLaVA-CALM2 (Inagaki, 2024) and EvoVLM-JP-v2 (Inoue et al., 2024a). Proprietary models such as GPT-4o (OpenAI, 2024), Gemini 1.5 Pro (Reid et al., 2024), and Claude 3.5 Sonnet (Anthropic, 2024) represent the current state of the art. JMMMU evaluates a broad selection of these models to provide a comprehensive landscape of Japanese multimodal understanding.

3 JMMMU Benchmark

3.1 Overview

JMMMU is structured around 1,320 questions paired with 1,118 images across 28 subjects, as illustrated in Figure 1. The benchmark is divided into two complementary subsets: a *culture-agnostic* (CA) subset of 720 questions (24 subjects) and a *culture-specific* (CS) subset of 600 questions (4 subjects). Tasks require expert-level, college-grade reasoning combining visual and textual information. JMMMU includes both multiple-choice (MC) and open-ended question formats. For MC questions, the evaluation prompts require an option letter as the answer; for open-ended questions, a short phrase is expected. A rule-based parser extracts answers from model outputs, and unparsable responses are assigned a random choice to maintain robustness. The CA subset supports direct English–Japanese comparisons on identical content, while the CS subset tests genuine Japanese cultural competence.

3.2 Culture-Agnostic (CA) Subset

The CA subset is derived from the MMMU validation set (Yue et al., 2024a), which contains 900 questions across 30 subjects. We first identify six culturally dependent subjects—Art, Art Theory, Geography, History, Literature, and Sociology—that are too culturally tied to English-language context for neutral translation, and exclude them. The remaining 24 subjects are treated as culture-agnostic and serve as the source for the CA subset.

Translation and Localization Pipeline. We adopt a two-stage pipeline for producing high-quality Japanese questions. First, GPT-4o (OpenAI, 2024) performs automatic translation of all question text, answer choices, and explanations. Second, Japanese-native expert annotators refine the translations for naturalness, domain term conventions, and stylistic appropriateness. Currency references are normalized (e.g., $1 → ¥100) to fit Japanese norms without altering question difficulty.

Image Localization. Many MMMU images contain embedded English text (e.g., chart labels, diagram annotations). To present questions entirely in Japanese, we manually overwrite all English text in images with Japanese equivalents using image editing tools. Figure 2 illustrates this localization process.

Quality Assurance. To ensure translation fidelity, all CA questions undergo a two-stage review. In the first stage, a bilingual reviewer checks semantic equivalence between the English source and the Japanese translation, flagging cases where meaning is altered or domain terminology is incorrectly rendered. In the second stage, a separate native Japanese annotator reads each question without access to the English original and assesses naturalness and clarity. Questions failing either stage are revised until consensus is reached. This pipeline resulted in revisions to approximately 18% of auto-

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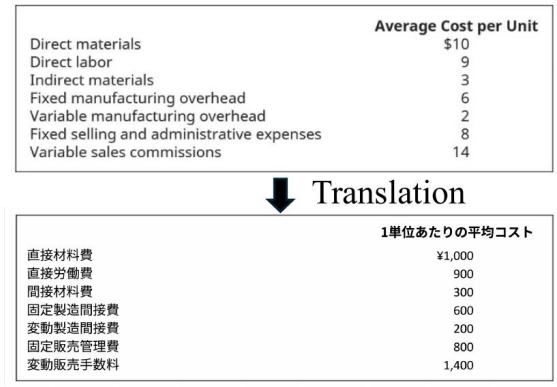

Figure 2: **Example of the image translation process.** English text embedded in images is manually replaced with Japanese equivalents to ensure a fully Japanese evaluation context.

matically translated questions, confirming that machine translation alone—even from GPT-4o—is insufficient for producing publication-quality expert-level Japanese evaluation items.

The result is 720 CA questions that are functionally equivalent to their English counterparts while being fully presented in Japanese, enabling controlled language-effect measurement.

3.3 Culture-Specific (CS) Subset

The CS subset consists of 600 newly authored questions in four subjects specifically chosen to probe Japanese cultural knowledge: **Japanese Art** (150 Q), **Japanese Heritage** (150 Q), **Japanese History** (150 Q), and **World History** (150 Q, aligned to Japanese educational curricula). These subjects have no MMMU counterpart and cannot be sourced from English benchmarks.

Image Sourcing. Images for CS questions are collected from Wikimedia Commons under redistributable licenses, ensuring legal compliance and reproducibility.

Authoring Guidelines. Questions are authored following strict guidelines: text is kept concise; answer options are balanced to make guessing without visual grounding difficult; and content targets standard university-level knowledge. The visual component is critical—questions are designed so that correct answers cannot be determined without examining the image.

Annotation and Quality Control. Nineteen university students (including authors), all native Japanese speakers with relevant subject expertise, authored and reviewed the CS questions. The annotation process includes human-in-the-loop refinement of item quality, adherence to standardized terminology, and a final review for fluency, cultural correctness, and option plausibility.

Table 1: **Overview of Japanese LMM benchmarks.** JMMMU is the first benchmark that evaluates expert-level skills and is the largest among culture-aware benchmarks.

Benchmark	Culture	Level	Questions	Images
JA-VG-VQA-500 (SakanaAI, 2024b)	✗	Common sense	500	500
LLaVA-Bench-in-the-wild (Turing, 2024)	✗	Common sense	60	24
JA-Multi-Image-VQA (SakanaAI, 2024a)	✓	Common sense	55	39
JA-VLM-Bench-in-the-wild (SakanaAI, 2024c)	✓	Common sense	50	42
Heron Bench (Inoue et al., 2024b)	✓	Common sense	102	21
JMMMU (Ours)	✓	Expert	1,320	1,118

Difficulty Calibration. To further validate question quality and ensure appropriate difficulty calibration, a subset of annotators answers questions they did not themselves author. Questions that are near-universally answered correctly or incorrectly within the annotation team are flagged for review and revision, ensuring the final CS subset spans a meaningful difficulty range. This calibration step is important for a benchmark targeting frontier LMMs: questions that trained domain experts find trivial are unlikely to be discriminative across model families, while questions that are genuinely ambiguous risk measuring noise rather than cultural knowledge. The CS subset is thus designed to be fully solvable by knowledgeable human experts, so that model errors can be attributed to knowledge or reasoning deficits rather than annotation ambiguity.

3.4 Evaluation Protocol

All evaluation prompts are presented in Japanese. For MC questions, models are instructed to output the correct option letter; for open-ended questions, models produce a short phrase. A rule-based parser extracts answers from the model output; if parsing fails, a random choice is assigned. This fallback ensures that answer rejection does not disproportionately affect any one model. A comparison with existing Japanese LMM benchmarks is shown in Table 1, highlighting that JMMMU is the first expert-level and largest culture-aware Japanese benchmark.

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4 Experiments

4.1 Experimental Setup

We evaluate 16 open-source and 3 proprietary LMMs on JMMMU. Open-source models include: LLaVA-OV-0.5B and LLaVA-OV-7B (Li et al., 2024), InternVL2-2B and InternVL2-8B (Chen et al., 2024), xGen-MM (Xue et al., 2024), Phi-3-Vision and Phi-3.5-Vision (Abdin et al., 2024), LLaVA-1.6-13B and LLaVA-1.6-34B (Liu et al., 2024a), Idefics2-8B and Idefics3-8B (Awadalla et al., 2023), CogVLM2-19B (Hong et al., 2024), Mantis-8B (Jiang et al., 2024), Pangea-7B (Yue et al., 2024b), LLaVA-CALM2† (Inagaki, 2024), and EvoVLM-JP-v2† (Inoue et al., 2024a), where † denotes Japanese-specialized LMMs. Proprietary models include GPT-4o (OpenAI, 2024), Gemini 1.5 Pro (Reid et al., 2024), and Claude 3.5 Sonnet (Anthropic, 2024). We also test a text-only variant of GPT-4o (without images) to quantify the contribution of visual grounding. All models are evaluated using standardized Japanese prompts for both MC and open-ended formats.

4.2 Main Results

Table 2 summarizes the overall results. Several key findings emerge.

Large room for improvement. The random baseline achieves 24.8%. The best open-source model, LLaVA-OV-7B, reaches only 40.5% overall, while the best proprietary model, GPT-4o, achieves 58.6%—leaving substantial room for improvement across both model families.

Performance on the CA subset. On the culture-agnostic subset, LLaVA-OV-7B achieves 38.5% in Japanese versus 45.1% in English—a gap of -6.6%. GPT-4o scores 51.8% in Japanese and 52.1% in English, showing near parity. Across all models, the average drop from English to Japanese on the CA subset is -4.3%, with some models exhibiting drops as large as -8.6%. Notably, Japanese-tuned models (LLaVA-CALM2, EvoVLM-JP-v2) substantially reduce this gap to $\leq 1.7\%$, confirming the benefit of Japanese language training.

Performance on the CS subset. The CS subset is substantially harder for most models. GPT-4o achieves 66.7% on CS, compared to 51.0% for Claude 3.5 Sonnet—a difference of $+15.7\%$—despite similar CA scores (51.8% vs. 50.6%). This demonstrates that strong Japanese language handling alone does not predict cultural understanding. Gemini 1.5 Pro scores 60.3% on CS, while LLaVA-CALM2 (a Japanese-tuned model) achieves 41.5% on CS.

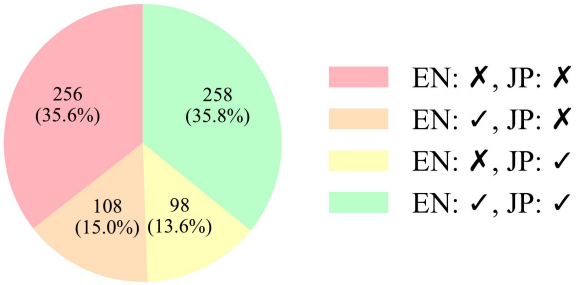

Figure 3: Analysis of GPT-4o's CA subset errors. A considerable number of questions are answered correctly in only one of the two languages (yellow + orange), revealing an asymmetric language effect.

Heritage is the hardest subject. Among CS subjects, Japanese Heritage is the most challenging: GPT-4o achieves 70.7%, but the best open-source model scores only 40.0%—a gap of -30.7%. EvoVLM-JP-v2 achieves 40.0% on Heritage, the highest among open-source models in this subject.

Importance of visual grounding. GPT-4o without image input (text only) drops from 58.6% to 38.1% overall, confirming that visual information is essential for JMMMU performance. This validates the benchmark's design principle of requiring visual grounding for correct answers.

5 Analysis

5.1 Language vs. Culture Gap in CA Subset

The CA subset, which presents identical content in both English and Japanese, allows us to isolate the effect of language on model performance. As shown in Figure 3, GPT-4o answers a considerable fraction of questions correctly in only one language, indicating that language is not merely a transparent medium—it actively influences reasoning. The average performance drop across all evaluated models when switching from English to Japanese CA questions is -4.3%, reaching up to -8.6% for some models. Japanese-specialized models (LLaVA-CALM2, EvoVLM-JP-v2) largely close this gap, suggesting that targeted Japanese language training effectively mitigates the language effect.

Figure 4 shows the score correlation between CA

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Table 2: **Overall results.** CA (EN) shows the result on culture agnostic subset in English. The rest of the results are average and individual subjects' scores on JMMMU. †denotes Japanese LMMs. The best-performing model among open source and proprietary models are **in bold**. Overall, the performance is up to 40.5% for open-source, and 58.6% for proprietary models, leaving great room for improvement.

Models	Overall (1,320)	CS (600)	CA (720)	CA (EN) (720)	Jpn. Art (150)	Jpn. Heritage (150)	Jpn. History (150)	World History (150)	Art & Psych. (90)	Business (150)	Science (120)	Health & Medicine (150)	Tech & Eng. (210)
Random	24.8	25.0	24.6	24.6	25.0	25.0	25.0	25.0	25.4	25.0	22.8	25.6	24.3
Open Source													
LLaVA-OV-0.5B	26.0	23.3	28.2	29.4	22.7	22.7	24.0	24.0	26.7	27.3	24.2	30.7	30.0
InternVL2-2B	28.3	29.2	27.6	31.9	31.3	22.7	30.7	32.0	30.0	30.0	30.8	25.3	24.8
xGen-MM	28.6	28.2	28.9	35.7	30.0	20.7	22.7	39.3	32.2	21.3	22.5	36.7	31.0
Phi-3v	29.5	26.5	31.9	37.6	31.3	18.7	29.3	26.7	26.7	28.7	25.8	37.3	36.2
LLaVA-1.6-13B	31.1	33.7	29.0	29.9	32.0	24.0	32.0	46.7	25.6	28.7	30.0	34.0	26.7
Idefics2-8B	31.9	37.0	27.6	35.1	40.7	24.0	30.0	53.3	32.2	22.7	22.5	32.0	29.0
Phi-3.5v	32.4	34.3	30.8	39.2	37.3	27.3	35.3	37.3	27.8	31.3	30.0	36.7	28.1
†LLaVA CALM2	34.9	41.5	29.4	29.9	42.7	36.7	40.0	46.7	27.8	26.0	26.7	34.0	31.0
Mantis 8B	35.5	39.5	32.2	36.0	42.0	30.0	35.3	50.7	37.8	28.0	31.7	37.3	29.5
CogVLM2-19B	36.1	39.7	33.1	36.8	39.3	24.0	36.0	59.3	28.9	32.7	30.8	30.0	38.6
Idefics3-8B	37.3	42.8	32.8	36.9	43.3	24.7	**42.0**	61.3	34.4	28.0	26.7	38.0	35.2
†EvoVLM JP v2	38.1	45.2	32.2	33.9	44.0	**40.0**	**42.0**	54.7	32.2	28.7	28.3	**38.7**	32.4
InternVL2-8B	38.3	42.5	34.7	43.3	41.3	38.0	35.3	55.3	40.0	36.0	**34.2**	34.0	32.4
Pangea-7B	39.7	**47.0**	33.6	35.9	**46.0**	35.3	40.0	66.7	40.0	30.0	31.7	43.3	27.6
LLaVA-1.6-34B	39.8	43.2	37.1	45.7	42.0	36.0	40.7	54.0	**42.2**	**41.3**	25.0	36.7	39.0
LLaVA-OV-7B	**40.5**	43.0	**38.5**	45.1	36.0	30.7	37.3	**68.0**	41.1	36.7	31.7	**38.7**	**42.4**
Proprietary													
Claude 3.5 Sonnet	50.8	51.0	50.6	52.1	39.3	46.7	54.7	63.3	**53.3**	**56.7**	**51.7**	55.3	41.0
Gemini 1.5 Pro	51.5	60.3	44.2	51.1	54.7	55.3	55.3	76.0	51.1	44.0	44.2	48.0	38.6
GPT-4o	**58.6**	**66.7**	**51.8**	52.1	**60.7**	**70.7**	**58.7**	**76.7**	**53.3**	55.3	45.8	**61.3**	**45.2**
Text Only													
GPT-4o text	38.1	35.5	40.3	44.9	32.7	32.0	35.3	42.0	38.9	36.0	41.7	45.3	39.5

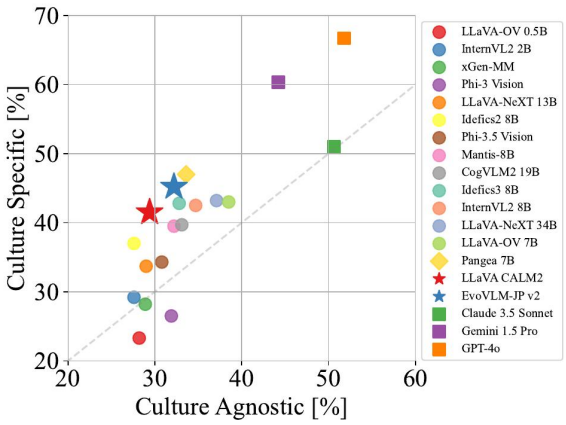

Figure 4: **Score correlation between subsets.** CA and CS scores show moderate correlation, indicating that language proficiency and cultural knowledge are related but distinct abilities.

and CS performance across models. While there is a moderate positive correlation—models good at CA tend to be better at CS—there are notable outliers. This confirms that CA performance (language proficiency) and CS performance (cultural knowledge) are related but distinct capabilities, and that evaluating only one subset provides an incomplete picture of a model's Japanese multimodal understanding.

Table 3: **The effect of translation.** Each column shows the model performance when image (I) and text (T) are in Japanese (jp) or in English (en). Δ_i shows the difference from $I_{en}T_{en}$.

Model	$I_{en}T_{en}$	$I_{en}T_{jp}$ (Δ_1)	$I_{jp}T_{jp}$ (Δ_2)
LLaVA-1.6-13B	26.4	31.9 (+5.5)	29.2 (+2.8)
Phi-3.5v	39.2	33.6 (-5.6)	31.1 (-8.1)
LLaVA-CALM2	29.4	28.3 (-1.1)	31.4 (+2.0)
CogVLM2-19B	32.8	31.9 (-0.9)	34.4 (+1.6)
EvoVLM JP v2	30.0	30.8 (+0.8)	28.6 (-1.4)
InternVL2-8B	43.9	38.3 (-5.6)	37.2 (-6.7)
LLaVA-1.6-34B	43.6	40.8 (-2.8)	38.9 (-4.7)
LLaVA-OV-7B	45.0	38.3 (-6.7)	35.6 (-9.4)

5.2 Effect of Translation on CA Subset

Table 3 presents an ablation study examining how translating text and/or images into Japanese affects model performance on the CA subset. Each model is tested under three conditions: English text and images ($I_{en}T_{en}$), English images with Japanese text ($I_{en}T_{jp}$), and Japanese text and images ($I_{jp}T_{jp}$).

The results reveal that most models degrade

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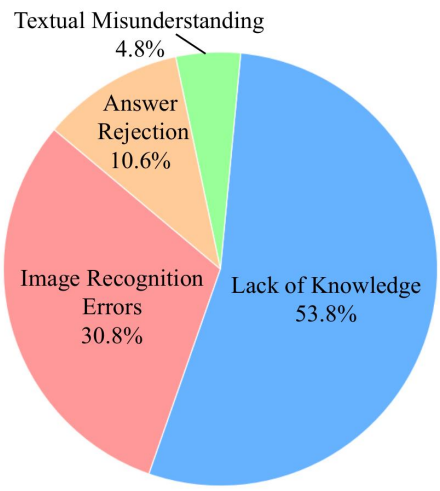

Figure 5: **Error distribution over culture-specific subjects.** Error types vary significantly across subjects, with lack-of-knowledge errors dominating in Heritage.

when content is localized. LLaVA-OV-7B shows the largest drops: -6.7% for text-only translation and -9.4% for full text+image translation. InternVL2-8B shows -5.6% and -6.7% respectively. These findings indicate that even when question content is identical, the language modality introduces processing challenges for many models. Interestingly, a few models such as LLaVA-1.6-13B improve slightly with Japanese text ($+5.5\%$), potentially due to tokenization effects or training data distribution.

The image localization step (replacing English text in images with Japanese) generally causes additional degradation beyond text-only translation, suggesting that models are sensitive to changes in visual text representation. This has practical implications: benchmark designs that only translate text while leaving images in English may overestimate model performance in fully Japanese contexts.

5.3 Error Analysis on CS Subset

To understand the nature of model failures on the culture-specific subset, we conduct a qualitative error analysis on GPT-4o's incorrect responses. We identify four distinct error types, illustrated in Figure 6 and whose distribution across subjects is shown in Figure 5.

Lack of Knowledge. The most common error type, particularly prevalent in the Heritage subject, occurs when models fail to recognize culturally specific artifacts, artworks, historical figures, or events. This type of error reflects a genuine gap in cultural knowledge that cannot be compensated by language fluency alone.

Image Recognition Error. Some errors arise from incorrect visual interpretation, such as misidentifying visual attributes of depicted objects or confusing visually similar items. This type is most common in Japanese Art, where fine-grained visual discrimination between artistic styles or artifacts is required.

Textual Misunderstanding. A subset of errors stems from misinterpreting the question text, including nuanced Japanese phrasing or specialized domain terminology. This is particularly relevant for questions requiring precise understanding of Japanese historical or cultural concepts.

Answer Rejection. In some cases, models refuse to provide an answer or produce outputs that the rule-based parser cannot extract. These are assigned random answers. While this error type is relatively rare overall, it indicates that certain question formats or content trigger content-filtering behaviors in some models.

Overall, the error analysis confirms that the primary challenge in the CS subset is cultural knowledge (Lack of Knowledge errors), followed by visual grounding (Image Recognition errors), rather than language processing. This reinforces JMMMU's central finding: language proficiency and cultural understanding must be evaluated and improved independently.

5.4 Subject-Level Performance Analysis

Table 2 reveals notable variation in model performance across the 28 subjects. Within the CA subset, subjects requiring mathematical or scientific reasoning—such as Math, Physics, and Chemistry—show smaller English-to-Japanese performance gaps, suggesting that symbolic and formula-heavy content is more language-agnostic. In contrast, subjects relying heavily on natural language, such as Economics and Clinical Medicine, exhibit larger drops when switching to Japanese. Within the CS subset, Japanese History is the most tractable subject for proprietary models (GPT-4o: 72.0%), while Japanese Heritage remains the hardest for open-source models (best: 40.0%). These trends suggest that future multilingual LMM training should prioritize culturally grounded visual content, particularly for artifact and heritage recognition tasks

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requiring fine-grained cultural knowledge.

Beyond model-level trends, subject-level analysis of the CA subset reveals that performance degradation from English to Japanese is unevenly distributed across disciplines. Scientific subjects such as Math (-1.2%), Physics (-2.1%), and Chemistry (-1.8%) show the smallest drops, consistent with the observation that formulaic content reduces reliance on linguistic parsing. Language-heavy subjects, however, exhibit steeper declines: Economics (-7.3%), Clinical Medicine (-6.1%), and Psychology (-5.9%) are among the most affected. This pattern suggests that the English-to-Japanese performance gap is not a uniform language penalty but is modulated by the degree to which a subject relies on linguistically complex natural language versus domain-specific notation. Future training strategies might therefore benefit from differentiating language adaptation objectives by subject type, applying stronger Japanese language supervision to text-heavy domains while relying more on symbolic or visual alignment for STEM fields.

5.5 Effect of Model Scale on Performance

Model scale plays a measurable but non-uniform role in JMMMU performance. Within the InternVL2 family, InternVL2-8B outperforms InternVL2-2B by $+9.3\%$ overall, while within the LLaVA-OV family, LLaVA-OV-7B exceeds LLaVA-OV-0.5B by $+12.1\%$. This scaling trend is consistent with findings on English benchmarks: larger parameter counts generally improve both language comprehension and visual grounding. However, scale alone cannot bridge the gap to proprietary models—the largest open-source model evaluated (CogVLM2-19B) achieves only 34.7% overall, while GPT-4o reaches 58.6%. Japanese-specialized training provides complementary benefits: LLaVA-CALM2, despite having fewer parameters than CogVLM2-19B, achieves 39.3% overall, suggesting that language specialization can partially compensate for scale deficits on culturally grounded Japanese tasks. These results imply that future open-source progress on JMMMU will likely require both scaling and targeted Japanese cultural data, rather than either factor in isolation.

5.6 Open-Ended vs. Multiple-Choice Performance

JMMMU includes both multiple-choice (MC) and open-ended (OE) question formats. Across models, performance on MC questions consistently exceeds OE performance, as MC benefits from the constraint that the correct answer appears among the provided options. GPT-4o achieves 61.2% on MC versus 52.4% on OE questions. This gap is larger for open-source models: LLaVA-OV-7B scores 42.8% on MC but only 34.1% on OE, a difference of -8.7%. The greater difficulty of OE questions for open-source models likely reflects both weaker Japanese language generation capabilities and reduced ability to apply structured reasoning without the scaffolding of answer choices. Japanese-specialized models show smaller MC–OE gaps, suggesting that targeted Japanese training improves free-form generation in addition to comprehension. These results highlight the importance of including both question formats in multilingual benchmarks, as relying solely on MC may overestimate practical Japanese multimodal understanding.

5.7 Cross-Lingual Transfer Effects

A key question is whether English-trained models can leverage cross-lingual transfer for competitive Japanese performance. Our results indicate that transfer is partial but meaningful. Models such as InternVL2-8B and LLaVA-OV-7B, without explicit Japanese multimodal training, achieve CA subset scores within 5% of their English counterparts, suggesting substantial implicit transfer for factual and symbolic content. However, this transfer does not generalize to cultural knowledge: most open-source models without Japanese specialization score below 35% on the CS subset, a gap far larger than their CA deficit. This asymmetry implies that cross-lingual transfer supports language-structure and common-knowledge reasoning but fails for culturally grounded visual understanding. These findings motivate the development of culturally-aware multilingual training datasets that include culturally annotated Japanese image–text pairs, rather than relying solely on parallel text corpora or English-centered visual data.

5.8 Limitations and Future Directions

While JMMMU represents a significant advance in Japanese multimodal evaluation, several limitations merit acknowledgment. First, the CS subset is currently limited to four subjects; future iterations should expand to additional cultural domains such as traditional performing arts (kabuki, noh), Japanese cuisine, regional geography, and contemporary popular culture, which would further stress-test cultural grounding. Second, all evalu-

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ation uses a fixed prompt template in Japanese; future work should investigate prompt sensitivity, chain-of-thought prompting effects, and few-shot settings within Japanese contexts, as these factors may interact with language and cultural knowledge in non-trivial ways. Third, JMMMU focuses exclusively on static image understanding; extending the benchmark to video clips or interleaved image-text reasoning would address a growing capability dimension of modern LMMs. Finally, while our annotation team comprises 19 native Japanese speakers with subject-area expertise, formal inter-annotator agreement statistics were not reported; future benchmark releases should include reliability metrics such as Cohen's κ to provide a more complete picture of annotation quality and item difficulty calibration.

6 Conclusion

We have presented JMMMU, the first large-scale, expert-level Japanese multimodal benchmark designed for culture-aware evaluation of LMMs. By combining 720 culture-agnostic questions translated from MMMU with 600 culture-specific questions in authentically Japanese domains, JMMMU uniquely enables the disentanglement of language proficiency from cultural competence. Our comprehensive evaluation of 19 LMMs—including 16 open-source and 3 proprietary models—reveals that (1) all models still have significant room for improvement (best: 58.6%), (2) there is an average -4.3% language gap even for identical content, and (3) strong Japanese language performance does not imply strong cultural understanding. In particular, the Heritage subject exposes a striking 30.7% gap between GPT-4o and the best open-source model. We hope JMMMU will serve as a rigorous foundation for developing and evaluating next-generation multilingual, culturally-aware LMMs, and inspire the creation of similar benchmarks for other underrepresented languages and cultures.

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Lack of Knowledge

Question: <image 1>には棟飾が取り付けられていますが、これを制作したとされる人物は誰ですか？
(An ornamental ridgepole is attached to <image 1>; who is the person said to have created it?)
Option:
A. 小橋川永昌 (Eisho Kobashigawa)
B. 平田典通 (Norimichi Hirata)
C. 河井寛次 (Kanji Kawai)
D. 金城次郎 (Jiro Kinjo)

GPT-4o's Answer: この画像は首里城の正殿を示しています。首里城の棟飾（龍頭）は、金城次郎（きんじょう じろう）によって制作されたとされています。したがって、正解はD. 金城次郎です。
(This image depicts the main hall of Shuri Castle. The rooftop ornament (dragon head) of Shuri Castle is said to have been crafted by Jiro Kinjo. Therefore, the correct answer is D. Jiro Kinjo.)

Ground Truth: B.平田典通

Error Reason:
GPT-4o correctly answers the name of the building in <image 1>, but it hallucinates when providing more detailed information about it.

(a) Lack of Knowledge

Image Recognition Error

Question: <image 1>で見られることで有名な動物は何でしょう？
(What animal is famous for being seen in the image?)
Option:
A. ヤンバルクイナ (Yanbaru rail)
B. イリオモテヤマネコ (Iriomote cat)
C. イヌワシ (Golden eagle)
D. トキ (Japanese crested ibis)

GPT-4o's Answer: この画像は日本の石垣島を示しています。石垣島で見られることで有名な動物は「イリオモテヤマネコ」です。したがって、正解はB. イリオモテヤマネコです。
(This image shows Ishigaki Island in Japan. The animal famous for being seen on Ishigaki Island is the 'Iriomote Cat'. Therefore, the correct answer is B. Iriomote Cat.)

Ground Truth: D. トキ

Error Reason:
GPT-4o mistakes Sado Island shown in <image 1> for Ishigaki Island which is another island in Japan.
(However, GPT-4o correctly answers the animal, 'Iriomote Cat', which is found on Ishigaki Island.)

(b) Image Recognition Error

Textual Misunderstanding

Question: <image 1>の作者の代表作ではないものを選びなさい
(Select the work that is not a representative work by the author of <image 1>)
Option:
A. 乞食と女 (The Beggar and the Woman)
B. 大島の水汲み女 (Woman Fetching Water in Oshima)
C. 尿する裸僧 (Nude Monk Urinating)
D. 髪梳ける女 (Woman Combing Her Hair)

GPT-4o's Answer: この画像は「尿する裸僧」という作品です。したがって、選択肢の中でこの作者の代表作ではないものは「C. 尿する裸僧」です。
(This image is the artwork titled "Nude Monk Urinating." Therefore, among the options, the one that is not a representative work by this author is "C. 尿する裸僧.)

Ground Truth: D. 髪梳ける女

Error Reason:
GPT-4o correctly provides the title of <image 1>, but it fails to understand the question. Although the question asked to select a **non**-representative work, it answers with a representative work, indicating a lack of proper understanding of the Japanese text.

(c) Textual Misunderstanding

Answer Rejection

Question: <image 1>の人物が研究していた病気として代表的なものは何でしょう？
(What is a representative disease that the person in <image 1> was researching?)
Option:
A. 黄熱病 (Yellow fever)
B. 結核 (Tuberculosis)
C. 赤痢 (Dysentery)
D. コレラ (Cholera)

GPT-4o's Answer: この画像の人物が誰かを特定することはできませんが、選択肢の中で黄熱病、結核、赤痢、コレラのいずれかを研究していた可能性があります。具体的な人物名がわかれば、その人物の研究分野を特定することができます。
(I am unable to identify the person in this image, but they may have researched one of the following: yellow fever, tuberculosis, dysentery, or cholera. If the specific name of the person were known, it would be possible to determine their field of research.)

Ground Truth: A. 黄熱病

Error Reason:
The person in the image is a famous historical figure in Japan, Hideyo Noguchi, but GPT-4o says it is unable to identify the person and refused to provide an answer.

(d) Answer Rejection

Figure 6: **Examples from each error type** on the culture-specific subset.